

Classification of Elements and Periodicity in Properties

The Periodic Table: Fundamental Concepts and Classification

The Periodic Table is considered the most important concept in chemistry. It provides a succinct organization of chemistry, demonstrating that chemical elements are not random but display trends and form families. An understanding of the Periodic Table is essential for anyone seeking to understand the fundamental building blocks of chemistry.

1. Historical Development of the Periodic Table

• Early Attempts at Classification:

- o By 1800, 31 elements were known, increasing to 63 by 1865, and currently, 114 elements are known, with efforts to synthesize more. The need for classification arose because studying each element and its innumerable compounds individually was very difficult.

- o Johann Dobereiner (early 1800s): First to consider trends among properties of elements. By 1829, he noted similarities in physical and chemical properties of "triads" (groups of three elements). In these triads, the middle element had an atomic weight approximately halfway between the other two, and its properties were also intermediate. Examples include Li, Na, K; Ca, Sr, Ba; and Cl, Br, I. However, this "law of triads" worked for only a few elements and was dismissed as coincidence.

- o A.E.B. de Chancourtois (1862): A French geologist who arranged known elements by increasing atomic weights on a cylindrical table to show periodic recurrence of properties, but this did not attract much attention.

- o John Alexander Newlands (1865): An English chemist who proposed the "law of octaves". He arranged elements in increasing order of atomic weights and observed that every eighth element had properties similar to the first, analogous to

musical octaves. This law was true only for elements up to calcium and was not widely accepted at the time, though he was later awarded the Davy Medal in 1887.

- Mendeleev's Periodic Law:

- o Developed by Dmitri Mendeleev (Russian) and Lothar Meyer (German), working independently. Both proposed in 1869 that arranging elements by increasing atomic weights revealed similarities in physical and chemical properties at regular intervals.

- o Lothar Meyer plotted physical properties (like atomic volume, melting point, boiling point) against atomic weight, observing a periodically repeated pattern with changing lengths. His table, developed by 1868, closely resembled the Modern Periodic Table, but Mendeleev is generally credited due to earlier publication.

- o Mendeleev published the Periodic Law for the first time, stating: "The properties of the elements are a periodic function of their atomic weights".

- o He arranged elements in horizontal rows and vertical columns by increasing atomic weights, ensuring elements with similar properties were in the same vertical column (group).

- o Mendeleev's system was more elaborate; he recognized the significance of periodicity and used a broader range of physical and chemical properties, especially similarities in empirical formulas and properties of compounds.

- o Boldness and Success: Mendeleev sometimes ignored the strict order of atomic weights, thinking measurements might be incorrect, to place elements with similar properties together. For example, iodine (lower atomic weight) was placed after tellurium (Group VI) in Group VII with fluorine, chlorine, and bromine due to property similarities. He also left several gaps for undiscovered elements, predicting their existence and

general physical properties. His predictions for Eka-aluminium (Gallium) and Eka-silicon (Germanium), including atomic weight, density, melting point, and formulas of oxides and chlorides, were later experimentally confirmed, making him and his table famous.

• Modern Periodic Law:

- o Mendeleev developed his table without knowledge of the internal structure of atoms.

- o In 1913, English physicist Henry Moseley observed regularities in X-ray spectra, showing that atomic number (Z) is a more fundamental property than atomic mass.

- o The Periodic Law was modified to the Modern Periodic Law: "The physical and chemical properties of the elements are periodic functions of their atomic numbers".

- o Atomic number equals nuclear charge (number of protons) or the number of electrons in a neutral atom. The Periodic Law is essentially a consequence of the periodic variation in electronic configurations, which determine properties.

11. Structure of the Modern Periodic Table

- The "long form" of the Periodic Table (Fig. 3.2 in sources) is the most convenient and widely used.

- Periods (Horizontal Rows): Mendeleev called these "series". There are seven periods. The period number corresponds to the highest principal quantum number (n) of elements in that period.

- o Period 1: 2 elements (filling 1s orbital).

- o Period 2: 8 elements (filling 2s and 2p orbitals).

- o Period 3: 8 elements (filling 3s and 3p orbitals).

- o Period 4: 18 elements (filling 4s, 3d, and 4p orbitals). This includes the 3d transition series.

- o Period 5: 18 elements (filling 5s, 4d, and 5p orbitals), similar to period 4.

- o Period 6: 32 elements (filling 6s, 4f, 5d, and 6p orbitals). This includes the 4f-inner transition series (lanthanoid series).
- o Period 7: Incomplete, theoretically would have 32 elements (filling 7s, 5f, 6d, and 7p orbitals). Includes most man-made radioactive elements and the 5f-inner transition series (actinoid series).
- Groups (Vertical Columns): Also called families. Elements in the same group have similar outer electronic configurations and thus similar properties. Groups are numbered from 1 to 18 according to IUPAC recommendations, replacing older notations.
- Lanthanoids and Actinoids: These 14 elements from both the sixth (4f) and seventh (5f) periods are placed in separate panels at the bottom of the table to maintain its structure and keep elements with similar properties in a single column. Glenn T. Seaborg's work, including the discovery of plutonium and other transuranium elements, led to the reconfiguration placing actinoids below lanthanoids.

III. Nomenclature of Elements with Atomic Numbers > 100

- Traditionally, discoverers named new elements, ratified by IUPAC, but this led to controversy (e.g., element 104).
- To avoid disputes, IUPAC recommended a systematic nomenclature for elements whose discovery is not yet proved and officially recognized. This system derives a temporary name directly from the atomic number using numerical roots for 0 and 1-9 (e.g., "un" for 1, "bi" for 2, "nil" for 0).
- The roots are put together in order of digits of the atomic number, and "ium" is added at the end. For example, element 120 is "unbinilium" (Ubn).
- Once discovery is proved, a permanent name and symbol are given by IUPAC vote, often reflecting the country of discovery or honoring a notable scientist. All elements up to 118 have official names.

IV. Types of Elements: S-, P-, D-, F- Blocks

Elements are classified into four blocks based on the type of atomic orbitals being filled with electrons.

• s-Block Elements:

- o Groups 1 (alkali metals, ns^1) and 2 (alkaline earth metals, ns^2).
- o All are reactive metals with low ionization enthalpies.
- o Readily lose outermost electrons to form $1+$ or $2+$ ions.
- o Metallic character and reactivity increase down the group.
- o Never found pure in nature due to high reactivity.
- o Compounds are predominantly ionic (except for lithium and beryllium).
- o Exception: Hydrogen, with one s-electron, is a special case and placed separately at the top of the Periodic Table.

• p-Block Elements:

- o Groups 13 to 18.
- o Together with s-block, they are called Representative Elements or Main Group Elements.
- o Outermost electronic configuration varies from ns^2np^1 to ns^2np^6 .
- o Noble gases (Group 18) have stable, completely filled valence shells (ns^2np^6) and exhibit very low chemical reactivity.
- o Halogens (Group 17) and Chalcogens (Group 16) are important non-metals with highly negative electron gain enthalpies, readily gaining electrons to achieve noble gas configuration.
- o Non-metallic character increases from left to right across a period; metallic character increases down a group.
- o Exception: Helium strictly belongs to the s-block ($1s^2$) but is placed in the p-block (Group 18) due to its filled valence shell and noble gas properties.

• d-Block Elements (Transition Elements):

- o Groups 3 to 12 in the center of the Periodic Table.
- o Characterized by the filling of inner d orbitals; general outer electronic configuration is $(n-1)d^{1-10}ns^{0-2}$.
- o All are metals.
- o Mostly form colored ions, exhibit variable valence (oxidation states), paramagnetism, and are often used as catalysts.
- o Zn, Cd, and Hg (with $(n-1)d^{10}ns^2$ configuration) do not show most transition element properties.
- o They bridge the chemically active s-block metals and less active Groups 13 and 14 elements.
- f-Block Elements (Inner-Transition Elements):
 - o Two rows at the bottom: Lanthanoids (Ce to Lu, Z=58-71) and Actinoids (Th to Lr, Z=90-103).
 - o Characterized by filling of f-orbitals; general outer electronic configuration is $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$.
 - o All are metals.
 - o Properties within each series are quite similar.
 - o Actinoids generally have more complicated chemistry due to a larger number of possible oxidation states and are radioactive. Elements after uranium are called Transuranium Elements.
- Metals, Non-metals, and Metalloids:
 - o Metals: Comprise over 78% of known elements, located on the left side of the Periodic Table. Usually solids at room temperature (exceptions: mercury is liquid; gallium and caesium have low melting points). Have high melting and boiling points, are good conductors of heat and electricity, and are malleable and ductile.
 - o Non-metals: Located at the top right of the Periodic Table. Usually solids or gases at room temperature with low melting and boiling points (exceptions: boron and carbon). Poor conductors of heat and electricity, most non-metallic solids are

brittle, neither malleable nor ductile.

- o **Metalloids (Semi-metals)**: Elements bordering the thick zig-zag line in Fig. 3.3 (e.g., silicon, germanium, arsenic, antimony, tellurium). They show properties characteristic of both metals and non-metals.

- o **Trends**: Metallic character increases down a group and decreases from left to right across a period. Non-metallic character increases from left to right across a period.

V. Periodic Trends in Properties of Elements

There are observable patterns in physical and chemical properties as one moves down a group or across a period. Periodicity is related to electronic configuration.

• A. Trends in Physical Properties:

o **Atomic Radius:**

- The size of an atom is difficult to measure precisely due to the diffused electron cloud. "Covalent radius" (half the bond distance in a covalent molecule) and "metallic radius" (half the internuclear distance in a metallic crystal) are used as estimates.

- **Across a Period (left to right)**: Generally decreases. This is because outer electrons are in the same valence shell, and the effective nuclear charge increases (due to more protons), leading to increased attraction of electrons to the nucleus.

- **Down a Group (top to bottom)**: Increases regularly. The principal quantum number (n) increases, meaning valence electrons are farther from the nucleus, and inner energy levels shield outer electrons from nuclear pull, increasing atomic size.

- Noble gas radii are usually not compared with covalent radii, but with van der Waals radii.

o **Ionic Radius:**

- **Cation (positive ion)**: Formed by losing an electron; smaller than its parent atom because it has fewer electrons while

nuclear charge remains the same, leading to stronger attraction.

□ **Anion (negative ion)**: Formed by gaining an electron; larger than its parent atom because added electrons increase repulsion among electrons and decrease effective nuclear charge.

□ **Isoelectronic Species**: Atoms and ions with the same number of electrons (e.g., O^{2-} , F^- , Na^+ , Mg^{2+} all have 10 electrons). Their radii differ due to different nuclear charges. The cation with greater positive charge has a smaller radius (stronger attraction), while the anion with greater negative charge has a larger radius (greater electron repulsion).

o Ionization Enthalpy ($\Delta_i H$):

□ **Definition**: Energy required to remove an electron from an isolated gaseous atom in its ground state. The first ionization enthalpy ($\Delta_i H_1$) removes the first electron, the second ($\Delta_i H_2$) removes the second, and so on.

□ Ionization enthalpies are always positive because energy is required to remove an electron. Successive ionization enthalpies increase ($\Delta_i H_2 > \Delta_i H_1$) as it's harder to remove an electron from a positively charged ion.

□ **Across a Period (left to right)**: Generally increases. Increasing nuclear charge outweighs the shielding effect of inner electrons, holding outer electrons more tightly. Maxima occur at noble gases (stable closed shells), minima at alkali metals (low ionization enthalpy, high reactivity).

□ **Down a Group (top to bottom)**: Generally decreases. The outermost electron is farther from the nucleus, and increased shielding from inner electrons outweighs increasing nuclear charge, requiring less energy for removal.

□ **Exceptions**:

□ **Boron vs. Beryllium**: Boron ($Z=5$) has a slightly lower first

ionization enthalpy than beryllium ($Z=4$). This is because beryllium removes a $2s$ electron, while boron removes a $2p$ electron. $2s$ electrons penetrate the nucleus more than $2p$ electrons, meaning the $2p$ electron in boron is more shielded and easier to remove.

□ Oxygen vs. Nitrogen: Oxygen ($Z=8$) has a lower first ionization enthalpy than nitrogen ($Z=7$). In nitrogen, three $2p$ electrons occupy different orbitals (Hund's rule), while in oxygen, two of the four $2p$ electrons must occupy the same orbital, leading to increased electron-electron repulsion, making it easier to remove one from oxygen.

□ Shielding/Screening Effect: Inner core electrons shield valence electrons from the full nuclear charge, reducing the effective nuclear charge experienced by valence electrons. Shielding is effective when inner shells are completely filled.

o Electron Gain Enthalpy ($\Delta_{eg}H$):

□ Definition: The enthalpy change when an electron is added to a neutral gaseous atom to form a negative ion. It measures the ease with which an atom forms an anion.

□ Can be exothermic (negative $\Delta_{eg}H$, energy released) or endothermic (positive $\Delta_{eg}H$, energy supplied).

□ Group 17 (halogens) have very high negative electron gain enthalpies because gaining an electron leads to stable noble gas configurations.

□ Noble gases have large positive electron gain enthalpies because an added electron must enter a higher principal quantum level, leading to an unstable configuration.

□ Across a Period (left to right): Becomes more negative (easier to add an electron). Effective nuclear charge increases, bringing the added electron closer to the nucleus.

□ Down a Group (top to bottom): Becomes less negative (harder to add an electron). Atom size increases, so the added

electron is farther from the nucleus.

□ Exceptions: Electron gain enthalpy of O or F is less negative than the succeeding elements (S or Cl). This is because the $n=2$ quantum level (O or F) is smaller, leading to significant repulsion from existing electrons, while $n=3$ (S or Cl) offers a larger region of space, reducing repulsion.

o Electronegativity:

□ Definition: A qualitative measure of an atom's ability in a chemical compound to attract shared electrons to itself. Unlike ionization enthalpy or electron gain enthalpy, it is not a measurable quantity.

□ Pauling scale is most widely used; fluorine is assigned an arbitrary value of 4.0 (highest electronegativity).

□ Across a Period (left to right): Generally increases. As atomic radius decreases, attraction between valence electrons and the nucleus increases.

□ Down a Group (top to bottom): Decreases. This is due to increasing atomic radii.

□ Relationship to Metallic/Non-metallic Properties:

Electronegativity is directly related to non-metallic properties (tendency to gain electrons) and inversely related to metallic properties. Increase in electronegativity across a period means increased non-metallic properties, while decrease down a group means decreased non-metallic properties (or increased metallic properties).

• B. Periodic Trends in Chemical Properties:

o Valence or Oxidation States:

□ Valence is a characteristic property, understood in terms of electronic configurations. For representative elements, valence is usually equal to the number of electrons in the outermost orbitals or eight minus that number.

□ Oxidation state is often used interchangeably with valence.

It's defined as the charge acquired by an atom based on electronegativity considerations from other atoms in a molecule. For example, in OF_2 , oxygen has an oxidation state of +2 because fluorine (more electronegative) is -1. In Na_2O , oxygen is -2 and sodium is +1.

- Many elements, especially transition elements and actinoids, exhibit variable valence.

o Anomalous Properties of Second Period Elements (Li to F):

- The first element of each group (Li, Be, B, C, N, O, F) differs significantly from other members of its group.

- Examples: Lithium and beryllium form compounds with pronounced covalent character, unlike other members of their groups which predominantly form ionic compounds.

- Diagonal Relationship: Lithium's behavior is more similar to magnesium (element of the following group), and beryllium's to aluminum. This similarity is called a diagonal relationship.

- Reasons for Anomalous Behavior:

- Small size.

- Large charge/radius ratio.

- High electronegativity.

- Limited Valence Orbitals: The first member of a group has only four valence orbitals (2s and 2p) for bonding, limiting maximum covalency to 4 (e.g., boron forms BF_4^-). Subsequent members have nine valence orbitals (3s, 3p, 3d) and can expand their valence shell (e.g., aluminum forms AlF_6^{3-}).

- Multiple Bonding: First members of p-block elements show a greater ability to form $p\pi-p\pi$ multiple bonds to themselves (e.g., $\text{C}=\text{C}$, $\text{N}=\text{N}$) and to other second period elements (e.g., $\text{C}=\text{O}$, $\text{C}\equiv\text{N}$).

o Chemical Reactivity:

- Across a Period: Reactivity tends to be high at the two extremes (Groups 1 and 17) and lowest in the center.

□ Left extreme (Alkali Metals): High reactivity due to ease of electron loss (low ionization enthalpy), forming cations.

□ Right extreme (Halogens): High reactivity due to ease of electron gain (high negative electron gain enthalpy), forming anions.

□ This relates to metallic (left) and non-metallic (right) character.

□ Oxides: Elements on the two extremes of a period combine easily with oxygen.

□ Extreme left: Most basic oxides (e.g., Na_2O).

□ Extreme right: Most acidic oxides (e.g., Cl_2O_7).

□ Center: Oxides are amphoteric (behave as acidic with bases and basic with acids, e.g., Al_2O_3) or neutral (no acidic/basic properties, e.g., CO).

□ Down a Group:

□ Representative Metals (e.g., alkali metals): Reactivity increases. This is generally due to a gradual decrease in ionization enthalpies (easier to lose electrons). Metallic character increases.

□ Non-metals (e.g., halogens): Reactivity decreases. This is generally due to a regular decrease in electron gain enthalpies (harder to gain electrons). Non-metallic character decreases.